CSE590 · MIDTERM PROGRESS

When and why does ApET work?

Understanding reconstruction-based visual token compression in LLaVA-1.5-7B

THE COST

Each image becomes 576 visual tokens, which dominate inference cost.

APET'S IDEA

Keep tokens that are hard to reconstruct linearly; merge the easy ones away.

OUR QUESTION

Does "hard to reconstruct" match what the model needs to answer?

Peng Chen · SBU ID 117702477 · October 2026

Three research questions; first-half goals met

RQ1 Diagnostics largely complete

How does reconstruction-based redundancy relate to a token's contribution to the answer, across tasks and layers?

RQ2 Comparisons done; attribution open

How much of ApET's effectiveness comes from error-based selection, and how much from merging?

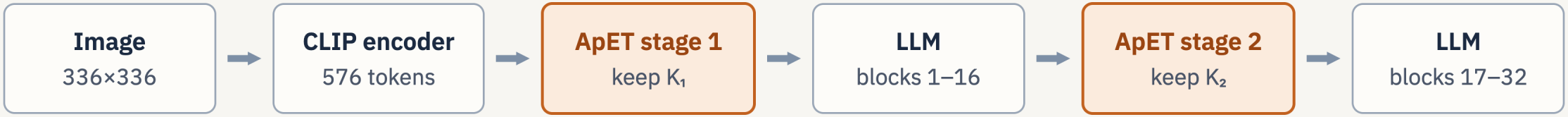
RQ3 Exploratory; none chosen

Can the analysis support a lightweight improvement, such as an adaptive budget?

FIRST-HALF OBJECTIVES · ALL MET

- Run native ApET and uncompressed LLaVA-1.5 in a reproducible environment
- Prepare GQA, MMStar and TextVQA subsets, baselines and scoring
- Instrument errors, selections, merges and answer responses

ApET keeps hard-to-reconstruct tokens and merges the rest



INSIDE EACH APET STAGE

1 · Pick a basis

10 tokens chosen by farthest-point sampling in feature space.

2 · Score and keep

Error E = how poorly a token is rebuilt from the basis. Keep the highest E.

3 · Merge the rest

Average each dropped token into its most similar kept token.

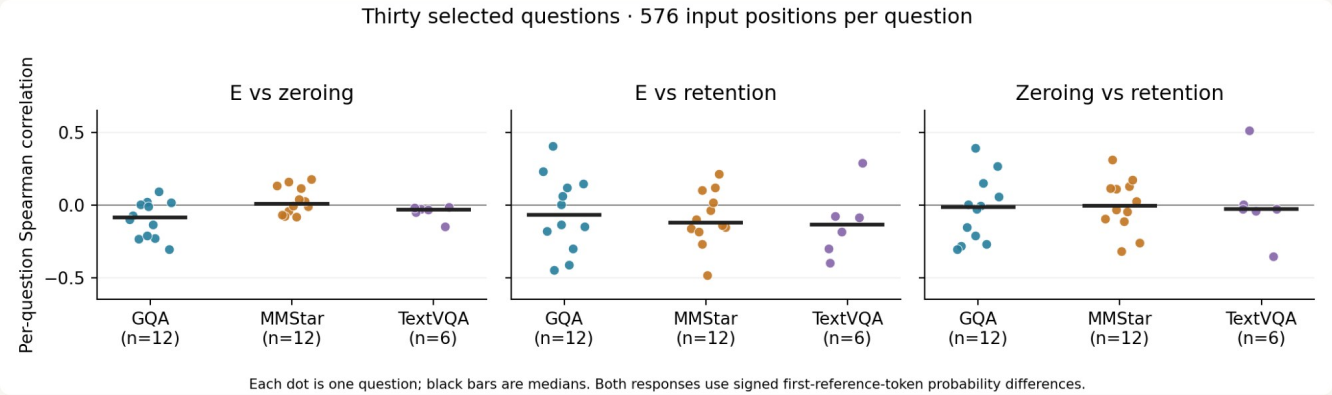
RQ1 · Does E rank tokens by their effect on the answer?

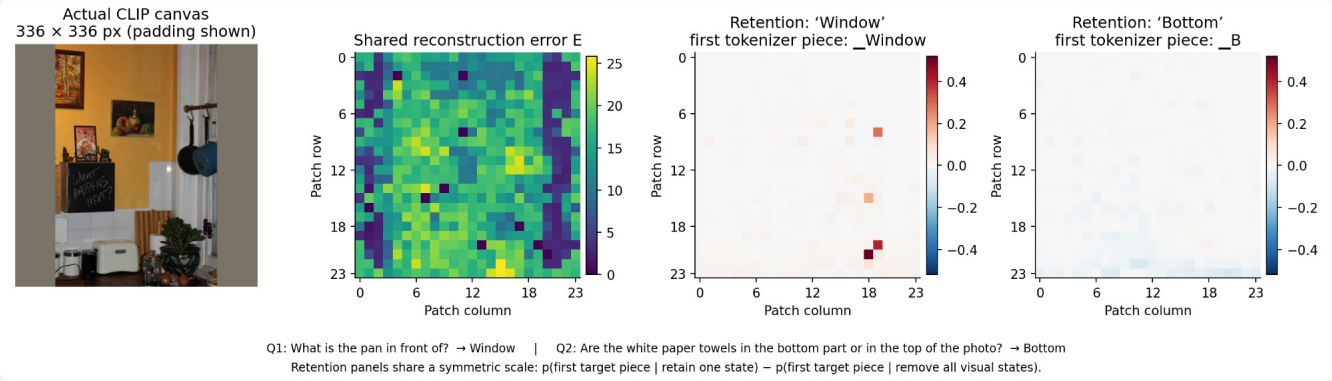
30-question diagnostic set; set one token to zero, or keep only one token.

RQ2 · Complete ApET vs. random pruning

Same budgets (avg. 64 or 192 tokens); GQA 800, MMStar 1,500, TextVQA 100; 3 seeds.

Finding 1: error E does not consistently rank tokens by their effect on the answer





−0.14 to +0.01

median correlation of E with answer effect, in every dataset (30 questions)

Single questions range from −0.49 to +0.40: not zero, but inconsistent.

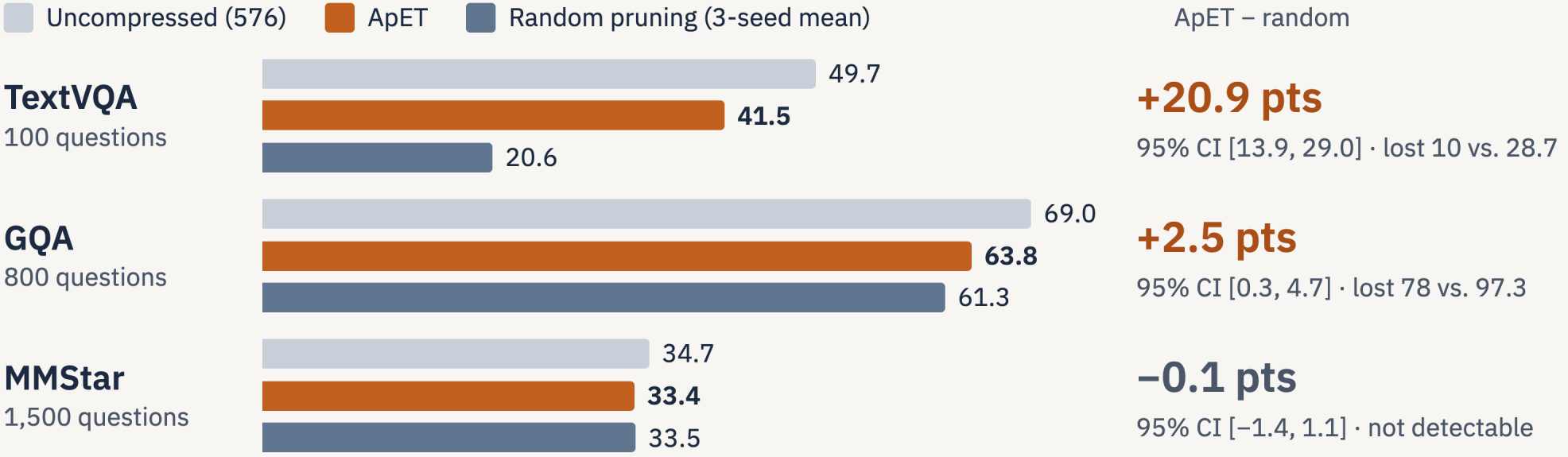
The two answer measures barely agree with each other either.

Bottom: one image, one E map, but each question depends on different tokens.

Answer effect = change in the first answer token's probability when one token is zeroed, or kept alone. · 4 / 9

ApET's advantage over random pruning is task-dependent

Accuracy (%) at an average budget of 64 visual tokens (schedule 96→32)



The puzzle: on its own, E does not consistently rank tokens by their effect on the answer (RQ1). So which part of ApET — selection, basis or merging — produces the gain?

"Lost" = answers correct without compression that became wrong. Same pattern, smaller, at 192 tokens. · 5 / 9

Why does ApET beat random? Two open hypotheses

HOW WE GOT HERE

Small component tests: random retention kept some answers ApET lost.

→

ApET's paper randomizes only the basis, never which tokens are kept.

→

So we compared at scale: ApET wins on average (Finding 2).

H1 · Selection

E-based selection keeps more useful tokens than random, e.g., by dropping easy padding or background, even if it ranks single tokens poorly.

Possible test: vary only selection; keep basis and merging fixed.

H2 · Representation preservation

Keeping basis representations and merging dropped tokens into kept ones preserves information that random pruning loses.

Possible test: merging on vs. off under the same selection.

Both may hold. The 30-question set cannot separate them: it was picked from ApET's own outcomes, so its 11 ApET losses are failures by construction.

RQ3 check, interpretation and validity

RQ3 FEASIBILITY CHECK

AUROC 0.41–0.57

Three image-level E features gave weak prediction of ApET losses; no clear gain over the base rate.

56 of 100 GQA images had both lost and kept questions under the same input.

Not adopted; question-aware policies remain open.

Worked

ApET clearly beats random on TextVQA, mainly via fewer losses; GQA retention close to the paper's.

Did not work

E does not consistently rank tokens by answer effect; image-level E did not predict failures.

Surprising failure

Merging changed "Brolio" to "Bottoli"; undoing one kept token's merge restored it.

Not yet met

Which component drives the gain (RQ2), and a supported RQ3 direction.

Validity checks: no-compression and fixed-feature replays matched reference outputs exactly; all 90 recomputed correlations matched; loss/gain counts consistent in all 27 conditions.

Feasibility check: five image-grouped folds on saved E values, no new GPU runs. · 7 / 9

Roadblocks and fixes

Independent issues, each with its own fix and status

Roadblock	What happened	Fix / action	Status
Integration	Kaggle package updates broke ApET/LLaVA compatibility.	Locked environment with recorded fixes.	Green: resolved
Run-to-run variation	Fresh compressed runs changed a few answers; replays from saved features were identical.	Diagnostic traced it to the merge step (likely order-dependent FP16 sums); keep first runs, repeat critical cases.	Yellow: managed, cause not isolated
Selection bias	The 30-question set was picked from ApET's own outcomes, so it favors other methods.	Use it only for RQ1 and case studies; compare components on unselected subsets.	Yellow: handled by design
Planning risk	Component attribution may stay unclear; no RQ3 candidate yet.	Report a well-controlled negative result.	Yellow: open

Merging uses CUDA scatter_add_; GPU additions can run in a different order each time, and FP16 addition is not exactly associative.

Next: does “easy to reconstruct” mean the model can still use it?

(a) Test ApET’s assumption

For low-E tokens, compare answers when a token is kept, merged, dropped, or replaced by its linear reconstruction.

(b) Look inside merging

Measure how averaged tokens differ from the originals; vary the merge rule with the kept tokens fixed.

Date	Milestone
Oct 30	Submit report and slides
Nov 6	Choose one comparison; write its protocol
Nov 13	Pilot on a small unselected subset
Nov 20	Main run at 96→32 (288→96 if feasible)
Nov 24	RQ3 go/no-go
Dec 15	Final presentation, report and code

SUCCESS CRITERION

One controlled comparison isolating a single ApET component on ≥100 unselected questions: matched budgets, 3 seeds, paired 95% intervals. Negative results count.

Stretch: an RQ3 modification, only if the analysis supports it.

Deprioritized: better basis selection (ApET’s own ablation shows FPS ≈ random basis). · 9 / 9